



TEAM HANDBOOK



TEAM HANDBOOK

Design creates culture

Culture shapes values

Values determine the future

2nd EDITION
2025

*Dedicated to our gracious mentors,
supportive parents, and our wonderful
students that inspire us every day.*

*Thank you for helping us build
such an impactful program.*

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Preface

In 2018, we took the plunge into the world of FIRST Robotics with the expectation to learn and have fun doing it. Back then, we set out to give our students a valuable experience tinkering with robots in a simple six week program with little thought about the impact we would generate in the years to come. Today, 7028 is a world class robotics team where incredibly talented kids can thrive under challenge, who may otherwise not have a school community to call home. This handbook is an encapsulation of our guiding principles that is intended to educate our members on how we operate. As 7028 continues to grow, we hope that these principles will serve each new member joining our ranks. If you are new to 7028, welcome! Although the words in this handbook are important, it's really your ideas, talent, and energy that will keep 7028 marching into the years ahead. Thank you for being here. Let's make this an experience you will never forget.

Is 7028 Right For Me?

7028 provides a very impactful but also very rigorous experience that does not suit every potential candidate. Oftentimes, the FTC program is a much better option for members who want less commitment, more control over design decisions, and/or less interaction with adults. Sharing any of the following ideas may indicate that FRC is probably not the right program for you, and that FTC may provide you a more valuable experience.

"I am only able to meet a few times per week, and will rarely stay past 5:00 pm."

"I want to participate in a program that is completely student-led, where adults are just supervising."

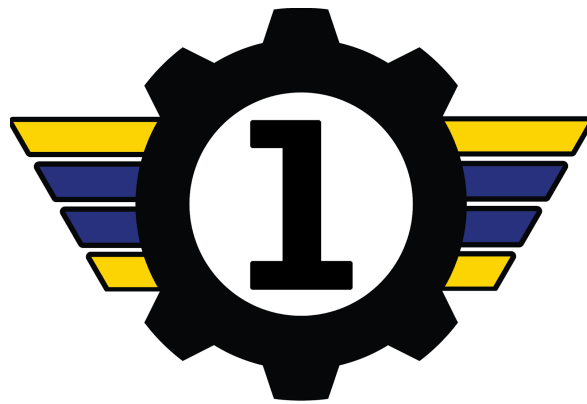
"I am uncomfortable competing with my teammates for my ideas to be adopted on the final product."

"I want a more relaxed experience where I can focus on building relationships with my peers rather than progressing the team."

"I am just looking for a casual after school activity to explore my interest in STEM."

"Collaborating with 40 other students sounds like something I would not enjoy."

Ultimately, it is up to you to decide what program is right for you. Our goal in offering both programs is to serve as many students as possible in their STEM journey.



Welcome to 7028

What is FIRST Robotics?

Founded in 1989 by inventor Dean Kamen, FIRST® (For Inspiration and Recognition of Science and Technology) is a global nonprofit organization that prepares young people for the future through a suite of inclusive, team-based robotics programs for ages 4-18 (PreK-12). FIRST's programs are suitable for schools or structured afterschool activities and receive support from a vast network of volunteers, educators, and sponsor/donors, including over 200 Fortune 500 companies.

FIRST LEGO LEAGUE		FIRST TECH CHALLENGE		FIRST ROBOTICS COMPETITION	
GRADES	AGES	GRADES	AGES	GRADES	AGES
PreK-8	4-16	7-12	12-18	9-12	14-18

STMA offers all levels of FIRST programming. 7028 is a First Robotics Competition (FRC) team, the highest and most prestigious level of the organization. FRC teams design, build, and program 125 pound robots in just six weeks to compete with other teams from around the world. No, it's not *Battlebots*, but we like to think that FRC is even more strategic and exciting than combat robotics. FRC robots compete on a volleyball sized field to score the most points by completing various tasks. Teams play each match in an "alliance" of three teams working together to win the match. The game completely changes every year.

Your First Year

Welcome to 7028!

So you've gone through the application process, registered for the team, and you're finally here at 7028. Congratulations and welcome!

Starting your journey on the team can be overwhelming with the scale of a First Robotics Competition team like 7028. This handbook is designed to ease your worries by providing a clear description of who we are and what you should expect. We want to make this the greatest collaborative STEM experience of your life, and that starts with a positive and productive first year.

A growth mindset is the key to becoming the next 7028 success story. Our greatest members are not necessarily the ones who bring the most skill or ability, but rather they bring a positive mindset and the will to learn and collaborate.

What should I work on?

Our students are free to pursue any area of the team they are passionate about, even if that area doesn't exist yet! There is no expectation that new members dive in and become super involved their first year. In fact, we recommend that every new student start with a goal of absorbing and learning as much as possible their first year. Be present and be ready to assist, but be careful about setting your sights too high as a rookie. Your 7028 career is years in the making. Take your time and enjoy every minute.

Part of being successful on 7028 is your ability to seek out and find new projects on your own. Checking in with mentors to receive tasks is a great way to start, but they may not always have something specific for you to do. This is where becoming a self-starter and seeking new tasks and new ways to add value independently is crucial.

What if I don't have the skills/ability?

Our coaches and mentors often say things like, "The build season is not the time to learn new skills." But if this is true, how can we expect our students to learn new things? Isn't that the entire point of the program? Yes, it is true that the build season does not include any formal education. Every day is a race to complete the next set of goals before the competition, leaving little room for formal education and training. Expecting students to have the skills to be successful while not actually teaching them anything is not the intention, and this is why we have our First Tech Challenge teams.

If you're looking to sharpen your robotics skills before the FRC season begins, joining a FIRST Tech Challenge (FTC) team provides the most effective experience. FTC allows students to develop a wide range of robotics skills on a smaller scale within a real competitive season. In the past, our team struggled to make preseason training both engaging and valuable, but we've realized that the best preparation for our students is for them to keep competing.

Can I be included the next time 7028 is deciding X?

Yes! There's no secret decision-making platoon (*well, maybe except for the Fortnite platoon*). No matter what project, you're already invited. All you have to do is either (1) Start working on it, or (2) Start talking to all the people who you think might be working on it already and find out how to best be valuable. You will be welcomed—there is no approval process or red tape involved. Quite the opposite—it's your job to insert yourself wherever you think you should be.

Mission and Values

Team Mission

To empower our students to grow into passionate and skilled STEM leaders through competitive robotics.

Team Values

- **Integrity**— Integrity is more than just telling the truth. Integrity is an all-encompassing idea that you should always do the right thing. This includes having discussions that are honest, making good decisions even when no one is looking, and always seeking the truth.
- **Self Advocacy**— Success is yours to own. It's your job to insert yourself wherever you think you should be, and to communicate your needs to coaches and mentors. Be an agent for change.
- **Growth Mindset**— Always look for ways to improve not only in skill but in attitude and relationships. Part of having a growth mindset is having humility for your current level of ability or lack thereof. Accept that you have a lifetime of untapped potential and that part of being in 7028 is unlocking that potential.
- **Excellence**— Commit to delivering high-quality work that meets or exceeds expectations. Be proud of the work you do by pushing yourself to achieve at a higher level. Don't settle for "just good enough".

- **Gracious Professionalism**— Gracious Professionalism is a way of doing things that encourages high quality work, emphasizes the value of others, and respects individuals and the community. Gracious professionals make valued contributions in a manner pleasing to others and to themselves. In the end, Gracious Professionalism is part of pursuing a meaningful life. When professionals use knowledge in a gracious manner and individuals act with integrity and sensitivity, everyone wins and society benefits.

Team Goals

- **Foster** a welcoming and safe STEM community that challenges our students to become their best self.
- **Grow** our students skills and knowledge through authentic, real-world, mentorship-based experiences.
- **Design and Build** competitive, world-championship worthy robots that we are proud of.
- **Impact** our community through STEM awareness, advocacy, and education.

“In FIRST, one of the most straightforward interpretations of gracious professionalism is that we learn and compete like crazy, but treat one another with respect and kindness in the process. We try to avoid leaving anyone feeling like they are losers. No chest thumping barbarian tough talk, but no sticky sweet platitudes either. Knowledge, pride and empathy comfortably blended.... In the long run, gracious professionalism is part of pursuing a meaningful life.”

-Woodie Flowers, FIRST Robotics Competition Founder

Policies

7028 is an equal opportunity organization

7028 and its members may not discriminate against any other member on the basis of immutable characteristics including but not limited to race, gender, sexual identity, or ability. This also includes individuals outside of the team at all team events such as competitions, outreach events, and fundraisers. We believe this sentiment is critical in creating an environment where everyone can thrive.

No member may use equipment they have not had safety training on

For the safety of our members, students and mentors may only use tools and equipment they have had formal safety training on in the past two years. Never use a machine you have not been trained on. Never use equipment you are unsure about. Never use equipment in a manner that is unsafe.

Students may never be alone with an adult.

For the safety of our students, no mentor, coach, parent, or any other adult may ever be left alone with a student during practices, competitions, or other team events. Students and adults alike are asked to always be aware of situations like these occurring and must be avoided.

7028 is an apolitical organization

While we believe it is absolutely critical for our students to freely express themselves on the team, political statements and symbols must not be associated with the 7028 brand. Political statements and symbols may not be present in any official team branding and/or marketing materials such as signs, pit, buttons, team shirts, handouts, or any other 7028 produced items.



What to Expect

Platoons, Commitment, and the Shop

What is a platoon?

Platoons are our on-brand way of organizing the team by discipline. Think of them as sub-teams or departments. Students indicated their platoon preference in the application form, but are not locked into one specific platoon for the entirety of the season. Consult the lead mentor of the platoon you wish to work with before collaborating with the group.

Design	Designs robots using CAD software
Manufacturing	Creates custom metal and plastic parts for robots
Build	Assembles robot parts according to designs
Control Systems	Wires and programs robot motors and sensors
Outreach	Plans and conducts community events Designs and presents for the impact award
Media and Marketing	Documents and promotes the team

How often do I need to attend?

The answer to this question greatly depends on what you expect to get out of the program. We regularly tell our students, “You get out of robotics what you put into it”. This essentially means that you get to decide how involved you are with the team.

Our competition season weekday meetings are only scheduled to 6:00 pm, but oftentimes groups of students will stick around an extra hour to work on their projects in order to meet deadlines. The reality is that progress will continue with or without you. It is up to you to decide if you want to be a casual member or fully dive into the robotics lifestyle, or anything in between.

While students occasionally choose to push themselves to work some extra hours at times when deadlines are approaching, for the most part working long nights for extended periods indicates a fundamental failure in planning or communication. If this happens at 7028, it's a sign that something needs to be reevaluated and corrected. We work hard to attract the most capable and intelligent students, and we want them to stick around and have a good balance between robotics and the rest of the important stuff in life. If you find yourself putting in long hours, or just generally feel like that balance is out of whack, be sure to raise the issue with your lead mentor or a coach.

We all love robotics. We love being in the shop with our friends. We simply love solving problems together. Being so passionate about anything in life is a huge win, and it is so fulfilling to see our students engaged and excited about STEM. But we seriously need to be careful about over-committing. Robotics should never be prioritized over your mental health or your school work. Coaches and mentors will never hold it against you for taking care of yourself over coming to robotics. Yes, we love what we do, but everyone needs breaks. Always keep in mind that this program is about YOU, not the work.

What do I need to know about the shop?

Our work area consists of five separate rooms (six including the Studio) that are designed for a variety of purposes. There are some amazing benefits to having such a large shop area but also some things to be aware of. Safety glasses are required for all shop areas except the CAD lab and the Studio. For safety reasons, never work alone in any of the shops. Never use equipment that you are not trained on or are unsure about using.

We share these spaces with hundreds of other STMA High School students who use them every day in their technology education classes. The shops do not belong to us; we are guests in the space. Because of this fact, it is important for all

members to follow our “Leave no Trace” philosophy, which essentially means clean up after yourself and leave the space the same as you found it.

Although they don’t necessarily belong to us, during robotics hours these shops are reserved for us and only us. No one outside of our team is allowed to be in the shops with us during these times. This is for the safety and security of the team.

7028 Facts That Matter

First Robotics Competition is a mentorship program.

The relationship between students and mentors is that of a partnership. Mentors and students alike must work together effectively to achieve our goals. Mentors are here to support students in achieving their goals, and students are here to grow.

7028 is supported by STMA Robotics.

STMA Robotics is a nonprofit organization that supports robotics programs in St. Michael-Albertville, including 7028. All funding for the team is donated by our sponsors and raised through team fundraisers. 7028 is not financially supported by the STMA school district and typically spends over \$100,000 per year.

7028 is not an STMA school sport/activity

7028 is run through STMA community education, not the high school itself. This means that students miss out some perks, like earning a letter. The benefit is that we have great freedom in how we decide to run our program. This is the way we prefer it at this time.

Flat Hierarchy

Assigned leadership roles are a great carrot to reward hard work and promote competition amongst members. But when you're a world class high school robotics team that's full of the most intelligent, innovative, and talented kids in the Midwest, telling them to sit back and do what they're told by another student diminishes their value and experience.

We believe that leadership is voluntary and does not require a title. That's why 7028 is flat. It's our shorthand way of saying that we don't have any student managers, and no student "reports to" any other student. We do have team captains, but these students hold no authority over others. Captains are simply role models for the team that exemplify our mission and values. This team is yours to thrive in. You have the power to share and execute your ideas. You have the power to pursue your interests. A flat structure removes every organizational barrier between your passion and the team reaching its goals.

Mentorship

It may be easy to assume that our students are free to run the program in any way they see fit and that the adults are just there to supervise, but this is far from what our team is about. In 7028, our amazing mentors partner with our students to help them grow their abilities and to provide structure and accountability. Our mentors have years of valuable FRC experience to help keep students on track with their goals, provide structure and support, and to fill in gaps. We agree with FIRST in that mentors are just as important to run a successful FRC program as the students.

Mentors are here to bring out the best in our students through inspiration, education, collaboration, and demonstration. That statement can be interpreted in many different ways, but it really depends on the circumstance. Mentors

completing tasks should never be prioritized over students completing those same tasks, but sometimes they need to fill in the gaps. It is totally normal and encouraged for mentors to work on a project alongside our students, whether that be helping turn a wrench, solder a wire, write some code, contribute ideas to discussions, or design a part. Your mentors are not passive observers and they are not your bosses. They are your partners in achieving your goals.



“Our mission is to create programs that give young people skills, confidence, and resilience to build a better world, and mentors are a key component to that in FIRST® Robotics Competition. Adult mentorship is part of what makes FIRST® so effective, and it’s been core to our programs from the very beginning. This is also true for the support given to teams by non-technical mentors in areas like logistics, business planning, fundraising, and outreach.”

“Part of being successful as an organization is knowing what you are and knowing what you aren’t. FIRST Robotics Competition is not a program where youth build a robot exclusively on their own to compete against another robot built exclusively by youth. It is a program where youth work both with each other and with adult mentors who help them learn new skills and grow as individuals.”

–Collin Fultz, FIRST Robotics Competition Senior Director

What We Expect From You

General Expectations

The only expectation we have for our students and mentors is that they follow our team values and policies. There is no expectation for you to perform to a certain level or commit a certain number of hours to the team.

When you are at an event wearing our logos, always be aware that you are representing our team. Our past and current members have worked very hard for many years to build our reputation to what it is today and it only takes one negative situation to bring that all down. As with anything, we expect you to follow our team values and policies while at events as well.

Failure and Risk

At 7028, you will never be reprimanded for making an honest mistake. It wouldn't make sense for us to operate that way. Providing the freedom to fail is an important trait of the team—we couldn't expect our students to grow if we also penalized them for errors. We can always repair the mistake or make up for it.

Screwing up is a great way to find out that your assumptions were wrong or that your model of the world was a little bit off. As long as you update your model and move forward with a better picture, you're doing it right. Look for ways to test your beliefs. Never be afraid to run an experiment or to collect more data.

It helps to make predictions and anticipate nasty outcomes. Ask yourself "what would I expect to see if I'm right?" Ask yourself "what would I expect to see if I'm wrong?" Then ask yourself "what do I see?" If something totally unexpected happens, try to figure out why. There are still some bad ways to fail. Repeating the same mistake over and over is one. Not listening to teammates or mentors before or after a failure is another. Never ignore the evidence; particularly when it says you're wrong.



Design Strategy

Decision Making

Making difficult decisions can be the most stressful, conflicting, and also the most important part of the design process. Are we going to use an arm or an elevator? How are we going to climb? Do we need to score in all possible scoring opportunities?

These are the types of questions that arise during the design process that will impact our season and our success. In 7028, decisions are not a vote, they are a consensus. Our *Design Strategy* does its best to tell us how to design the robot, but it is up to you to interpret its wisdom and apply it to the robot. When making decisions of any scale, consider these three questions:

1. **Does this decision align with our Design Strategy?**
2. **Does this decision align with our team and season goals?**
3. **Are the implications of this decision within our abilities?**

“Einstein Ready”

The robots we make must be designed to win the World Championship. World Championship winning robots play the game using the absolute best strategy while also having the ability to execute all game actions. So what do these robots have other than the best strategy?- they prioritize durability over consistency, and consistency over speed. World Champion robots never break, execute actions with near-perfect success rates, and also execute those actions as fast as possible.

Durability -> Consistency -> Speed

Black & White Thinking

Black and white thinking is a cognitive distortion where one views situations in only two categories without acknowledging nuance or gray areas. As much as we love to think that hard and fast rules about design will lead to predictable and consistent outcomes, that is just not the case in robotics, or engineering in general.

Yes, it is important to favor certain characteristics and ideals, but our success thrives in the gray area. Everyone would agree that it would be ideal to check every single box in the *favor* columns, but with limited resources compromises may need to be made.

7028 Design Strategy

Design

Required

Robot fully designed in CAD
Structurally stable and durable
Designed in inches to the 0.000
Full scope robot for regional 1
Nothing as an “afterthought”
Lighten the load for automation

Favor

Simple and robust
Proven designs
Low center of mass
Large acquisition zone
Hard contact with soft game pieces
Soft contact with hard game pieces
Easy maintenance

Avoid

Actuating claws/grabbers
Relying on human player skill
Game piece dead zones
Tight margins on perimeter and weight
Pneumatics
Over- complexity
Moving parts that share space

Build & Manufacturing

Required

- Backup parts
- Loctite on all threaded fasteners
- Follow manufacturer instructions for assembling COTS parts
- Bumpers changeable in under 2 minutes

Favor

- Rivets for connecting thin material
- Snap rings for retaining shafts
- Button head cap screws
- Consistent hardware types

Avoid

- Hand cut/drilled holes
- 1/8" aluminum

Control Systems

<i>Required</i>	
Recently tested, “MK” brand batteries <2 years old	
Soldered battery connectors	
Zip-tied battery connectors	
Anderson connectors for motors	
Wagos for all CAN and small gauge	
Ferrules for all push button connectors	
Mini power modules for branch circuits	
Smart tug all wires	
Motion profiling for position	
Closed loop for rotation	
LEDs to communicate robot state	
<i>Favor</i>	<i>Avoid</i>
Practical automation	Non-locking connectors
Encoders for things that move	Exposed/vulnerable wires
Limit switches and hard stops of limited range of motion systems	
Horizontally mounted battery	



History

2025 Reefscape



Torrent

Coaches

Ben Wandmacher
Zack Osowski

Captains

Grace Kirk
Kylie Lukkar
Tatum Lallak
Will Spaulding

Lake Superior Regional

Week 1

Qualifications	4-5 record	Rank 20/54
Playoffs	1st pick of alliance #3 (1732, 7028, 8122)	3-2

Granite City Regional

Week 6

Qualifications	8-2 record	Rank 4/48
Playoffs	#3 Alliance captain (7028, 7257, 7530)	4-3
Awards	Engineering Inspiration Regional Finalists	

Curie Division**World Championship**

Qualifications	9-1 record	Rank 3/76
Playoffs	#3 Alliance captain (7028, 27, 3061, 2847)	1-2

Minnesota State High School League Championship

Qualifications	6-1 record	Rank 6/36
Playoffs	#3 Alliance captain (7028, 2491, 4728)	1-2

2024 Crescendo



Tempo

Coaches

Kelly Willette
Kris Rue
Zack Osowski

Captains

Dylan Rattai
Sydney Gasser

Great Northern Regional

Week 2

Qualifications	3-6 record	Rank 45/56
Playoffs	1st pick of alliance #4 (4198, 7028, 2508)	0-2
Awards	Autonomous Award	

Granite City Regional

Week 6

Qualifications	9-1 record	Rank 3/54
Playoffs	1st pick of alliance #1 (3276, 7028, 6045)	5-0
Awards	Regional Winners Team Sustainability Award	

Curie Division

World Championship

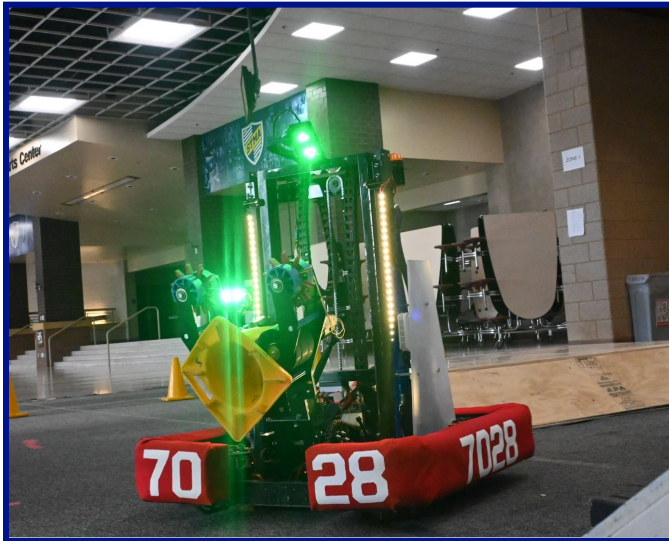
Qualifications	7-3 record	Rank 20/74
Playoffs	2nd pick of alliance #8 (2590, 4476, 7028, 190)	6-1
Awards	Championship Division Winner	

Einstein Field

World Championship

Match 3	2590	7028	4476	9483	604	78	138	148
Match 6	2590	7028	4476	5813	1477	3061	141	140
Match 9	6328	9072	4481	4476	7028	2590	125	119

2023 Charged Up



Orion

Coaches

Kris Rue
Roger Bovee
Zack Osowski

Captain

Audie Wadmacher

Great Northern Regional

Week 2

Qualifications	9-1 record	Rank 3/53
Playoffs	Captain of alliance #2 (7028, 876, 3630)	1-2
Awards	Innovation in Control Award	

Minnesota North Star Regional at La Crosse

Week 4

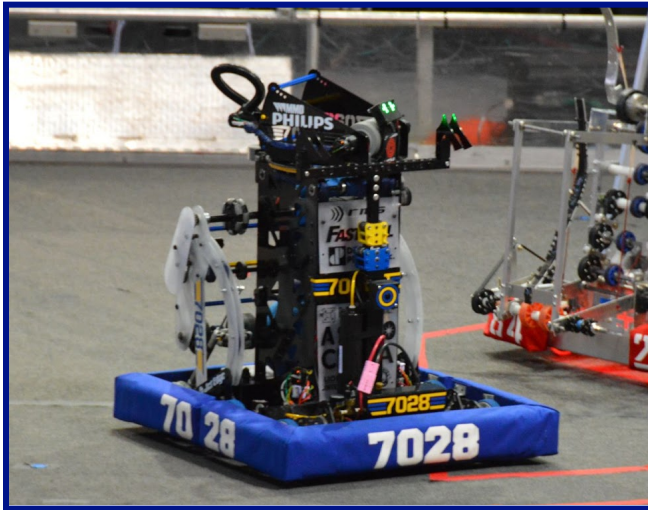
Qualifications	7-6 record	Rank 16/39
Playoffs	1st pick of alliance #7 (3082, 7028, 4198)	2-2
Awards	Autonomous Award	

Minnesota State Championship

MSHSL

Qualifications	2-6 record	Rank 30/36
Playoffs	-	-
Awards	-	

2022- Rapid React



Onyx

Coaches

Kris Rue
Roger Bovee
Zack Osowski

Captain

Matthew Bartos

Great Northern Regional

Week 4

Qualifications	7-3	Rank 4/49
Playoffs	1st pick of alliance #2 (5913, 7028, 2508)	1-2
Awards	Innovation in Control Award	

Minnesota 10,000 Lakes Regional

Week 6

Qualifications	8-2	Rank 3/55
Playoffs	1st pick of alliance #2 (2052, 7028, 3082)	5-2
Awards	Regional Finalists Autonomous Award	

Minnesota State Championship

MSHSL

Qualifications	4-4 record	Rank 11/36
Playoffs	1st pick of alliance #4 (2502, 7028, 2883)	0-2
Awards	-	

2020- Infinite Recharge



Wilson

Coaches

Kris Rue
Roger Bovee
Zack Osowski

Captain

Sam Draeger

Great Northern Regional

Week 1

Qualifications	6-3	Rank 13/60
Playoffs	1st pick of alliance #7 (4539, 7028, 3297)	0-2
Awards	-	

2019- Deep Space



Ozzy

Coaches

Kris Rue
Roger Bovee
Zack Osowski

Captain

Wyatt McCarthy

Northern Lights Regional

Week 2

Qualifications	5-3-1	Rank 11/60
Playoffs	1st pick of alliance #5 (6132, 7028, 4593)	1-2
Awards	-	

Minnesota 10,000 Lakes Regional

Week 5

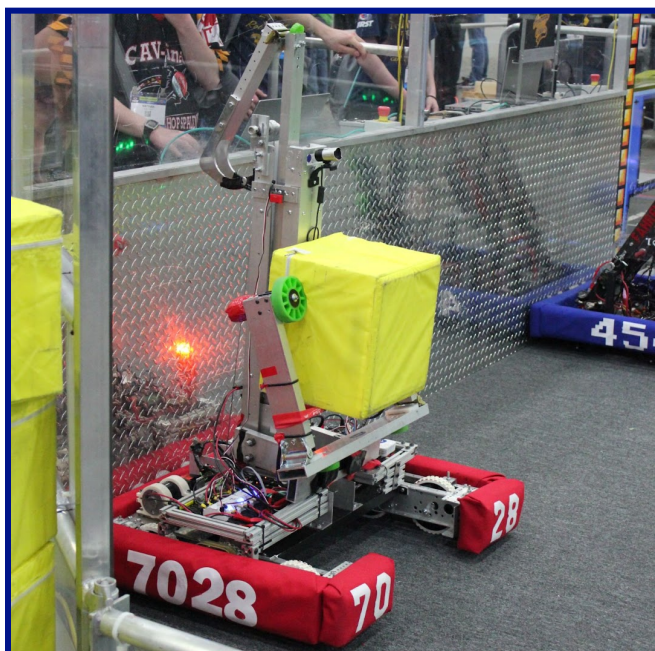
Qualifications	7-2	Rank 5/63
Playoffs	1st pick of alliance #2 (2987, 7028, 4536)	2-3
Awards	Creativity Award	

Minnesota State Championship

MSHSL

Qualifications	6-3 record	Rank 3/36
Playoffs	Captain of alliance #3 (7028, 4607, 1816)	0-2
Awards	-	

2018- Power Up



Phyllis

Coaches

Kris Rue
Roger Bovee

Captain

Wyatt McCarthy

Lake Superior Regional

Week 2

Qualifications	3-6	Rank 41/62
Playoffs	-	-
Awards	Rookie All Star Award Highest Rookie Seed	

Carson Division

World Championship

Qualifications	4-6	Rank 57/68
Playoffs	-	-
Awards	-	



Epilogue

What is 7028 Not Good At?

The design of the team has some downsides. We usually think they're worth the cost, but it's worth noting that there are a number of things we wish we were better at:

- Helping new students find their way. We wrote this handbook to help, but a book can only go so far.
- Disseminating detailed information across platoons.
- Finding and recruiting mentors and students in completely new disciplines (e.g., business and art).
- We miss out on recruiting talented students who prefer to work within a more casual structure. Again, this comes with the territory and isn't something we should change, but it's worth recognizing as a self-imposed limitation.
- Transferring skills and knowledge from veteran members to rookies.

What is 7028 Not Trying to Be?

7028 is a lot of things, and what it exactly is will change over time as the team grows. These are some things that the team is not and will never strive to be that may not be so obvious:

- A casual, stress free, and simple after school STEM program
- An internship-like program where students are just here to observe mentors
- A student-ran organization

What Happens When All This Doesn't Work?

Sometimes, the philosophy and methods outlined in this handbook don't match perfectly with how things are going day to day. But we're confident that even when problems persist for a while, 7028 roots them out. As you see it, are there areas of the team in which the ideals in this book are realized more fully than others? What should we do about that? Are those differences a good thing? What would you change? This handbook describes the goals we believe in. If you find yourself in a group or project that you feel isn't meeting these goals, be an agent of change. Help bring the group around. Talk about these goals with the team and/or others.

Where will you take us?

7028 will be a different team just a few years from now than it is today because you are going to change it for the better. We can't wait to see where you take us. The robots, media, and outreach experiences that you decide to create for the team are the things that will define us into the future. Whether it's a new mechanical design, a control systems innovation, an impact presentation, a robot reveal video, a new manufacturing process, a quality control system, a scouting app, a laser engraved breadstick, a new emoji, or (more likely) something nobody's thought of yet—we can't wait to see what kind of future you choose to build for yourself at 7028.

Glossary of Terms

FIRST— For Inspiration and Recognition of Science and Technology - a global nonprofit organization that prepares young people for the future through a suite of life-changing youth robotics programs that build skills, confidence, and resilience.

FLL— FIRST Lego League - Introduces science, technology, engineering, and math (STEM) to children ages 4-16 through fun, exciting hands-on learning.

FTC— FIRST Tech Challenge - Grades 7-12 are challenged to design, build, program, and operate robots to compete in a head-to-head challenge in an alliance format.

FRC— FIRST Robotics Competition - Combining the excitement of sport with the rigors of science and technology. The ultimate Sport for the Mind inspiring High-school students.

Mentor— An adult of at least 21 years old who is committed to mentoring the students of 7028. Mentors share our team values and mission in creating valuable experiences for our members. Mentors are background-checked with the STMA school district.

Lead Mentor— A veteran mentor assigned to oversee a specific platoon on the team. Each platoon has exactly one lead mentor.

Coach— An employee of STMA school district that leads the team.

Student— A 9-12th grade person who has been accepted to join 7028 as part of the application process.

Member— A coach, mentor, or student.

Odometry— Using sensors on the robot to create an estimate of the pose of the robot on the field.

KOP— Kit of Parts - the collection of items listed on the Kickoff Kit checklists, distributed to the team via FIRST Choice, or paid for completely (except shipping) with a Product Donation Voucher (PDV).

FMS— Field Management System - the electronics core responsible for sensing and controlling the FIRST Robotics Competition field.

FTA— FIRST Technical Advisor - FRC volunteer position that is responsible for ensuring FIRST Robotics Competition events run smoothly, safely, and in accordance with FIRST requirements, and ensuring a high-quality experience for all event participants and teams.

Build Season— The part of the competition season that starts at kickoff and ends when regionals begin

Competition Season— The season where a new game is introduced and robots are built and competed with, including the Build Season.

Preseason— The part of the year where resources are dumped into training members, maintenance and competing in offseason events with the previous year's robot, and completing infrastructure projects.

Offseason Event— An event that happens between competition seasons that is not an official FIRST event.

Regional— An official FIRST event with 50+ competing teams over three days.

Worlds/Champs— The FRC World Championship hosted in Houston, TX. A competition between the best 600 robot teams in the world.

Platoon— A group of students who share specific team responsibilities.

Control Systems (Platoon)— The platoon that programs and installs the electronic components of the robots.

Build (Platoon)— The platoon that assembles robots according to designs.

Manufacturing (Platoon)— The platoon that makes custom parts for robots in-house.

Design (Platoon)— The platoon that designs robots in 3D CAD software

Outreach (Platoon)— The platoon that plans, organizes, and carries out team outreach events. They also create and submit for the Impact Award.

Media and Marketing (Platoon)— The platoon that creates photo/video media and handles team social media.

Fortnite (Platoon?)— A mysterious, exclusive platoon that may or may not actually exist.

Sauce— Most often refers to robot code, but can also refer to any type of liquid used in the shop.

Meat— Power. Used in reference to motor speed, usually in game piece shooting applications.

Baby Bird— To pick up a game piece directly from the human player station without it touching the ground.

Yard Sale— When a robot and/or collection of game pieces is in the way and prevents teammates from reaching a position on the field.

DNP— “Do Not Pick”, a category of robots that have been identified as not to be picked in alliance selection.

Brown-Out— When a robot’s battery is low and voltage is dropping below what is required to operate properly.

Wilson Shakes— A recurring behavior seen in robots that shake back and forth while aiming. This is not necessarily a problem and is a result of aggressive PID tuning. Named after the 2020 robot, Wilson.

Goober— A person (usually a team member) whose behavior is out of line with team values and/or policies.

Sick— A synonym for “cool”. Can also describe a robot that is “ill”, meaning that it intakes and outputs from the same end, just like a sick person.

GCR— Granite City Regional held in St. Cloud Minnesota.

GNR— Great Northern Regional held in Grand Forks North Dakota.

IRI— Indiana Robotics Invitational offseason event.

MMR— Minnie Mini Regional offseason event.

Week 0— A scrimmage type event that occurs in the short time between the build season and the competition season.

OPR— Offensive Points Rating. A statistic calculated by the Blue Alliance that roughly estimates a team’s average points contribution during a match.

EPA— Expected Points Added. Similar to OPR but calculated by Statbotics.

Statbotics— A website that runs statistical information for all FRC teams.

Chief Delphi— The premiere FRC online forum.

Negative OPR— A reference to OPR, indicates a member’s behavior that is taking away productivity or progress for the whole team or group.

PID— Proportional, Integral, Derivative. A positional control system that affects the way motors move.

Forbidden Jelly— Red and tacky grease.

Nacho Supplies— Heat guns and other tools that produce heat.

Blue Alliance— A website/app that displays data for FRC events.

Ring Toss— Game pieces getting stuck/lodged on a robot.

Superalliance— 7028's in-house scouting app.

Slack— The main communication tool used by 7028.

Impact Award— The most prestigious award in FIRST, given to one team per regional that honors the team that best represents a model for other teams to emulate and best embodies the mission of FIRST.

COTS— Commercial, Off-The-Shelf product. Complete products designed and sold for FRC that are not made in-house like swerve modules, motors, and mechanisms.

Cheddar— Game pieces that are on the field, free for the taking.

Cheesy— FRC team 254, The Cheesy Poofs. Historically the most competitively successful FRC team of all time.

